FEEDBACK

YOUR SUGGESTIONS

THANKS!

The 'Top 20 Android Project Ideas' under Software Projects on *www.electronicsforu.com* has some really great articles!

Ajith Parma
Through email

EFY. Thanks for the feedback!

REMOTE-CONTROLLED SMARTFAN

This is regarding 'Remote Controlled Smartfan Using AT89C2051' DIY article published in December 2016 issue. I would like to thank the author for designing an interesting and useful circuit and EFY for publishing the same.

Resistors R6 through R9 are mentioned as 100E/2W each. These will be used in series with the fan/light, which will carry full-load current at full speed or brightness. Considering the load as an average of 75W (say), series resistors should be much more than 2W. A tapped resistor (wirewound) with higher wattage is OK.

Resistor network RNW1 could have been replaced with three 10-kilo-ohm, 1/4W resistors, which is economical and saves space. Moreover, ULN2003 could have been used in place of the three transistors to make the design compact.

M. Hyder Hossain
Through email

The author Pamarthi Kanakaraj replies:
In the circuit, the resistors are completely bypassed (so there is no voltage drop) when the fan is at full speed. In my prototype, I have used a tapped resistor as mentioned in the article.

Three 10-kilo-ohm resistors can be used for compactness as suggested by you. ULN2003 is made up of seven transistors (an array of seven npn Darlington transistors). But I used only three. Moreover, ULN2003 is costlier as compared to three BC547 transistors.

Corrections

In 'IoT-Enabled Air pollution Meter' DIY article published in January issue, 5V for relay operating voltage is not specified and the freewheeling diode across the relay is missing.

Tapan Mojidra
Through email

EFY. Thanks for pointing out the mistake!

In Q&A section of January issue, the last point in the answer to Q4 is wrong. Arduino Uno has 20 input/output pins—14 digital pins and six analogue pins. So there is no difference between the two as far as the number of input/output pins is concerned.

P. Rakesh, Tapan Mojidra and Ayush Khandelwal
Through email

EFY. Thanks for pointing out the mistake!

IR SENSOR BASED POWER SAVER

I constructed the infrared sensor based power saver based on the DIY article of the same name, published in March 2016 issue. It is in running condition but I have a doubt. Resistor R3 (220-kilo-ohm) caused too much voltage drop. So I removed potmeter VR1 (1-mega-ohm) and resistor R3, and replaced these with a 2.2-kilo-ohm resistor.

Muhammad Farhan
Through email

The author Fayaz Hassan replies:
I am happy that someone tried out the circuit and it is working fine. The main purpose of VR1 and R3 is to slowly charge capacitor C3 (1000µF), so that it increases time delay of the PIR or infrared sensor, which has a limited triggering time. If C3 has more internal leakage than the charge input through VR1 and R3, then C3 will never get charged and output of IC1 (NE555) will never change. So use a new and good quality capacitor with at least 25V rating. By changing the charging resistance, only relay hold time will vary and the circuit will function normally.

❑ Can 'Infrared Sensor Based Power Saver' project automatically switch on/off light and fan? Will the project work if two or three people enter the room while one goes out of the room?

Ayan Bhakta
Through email

The author Fayaz Hassan replies:
Yes, it will work. The circuit does not count the number of people. The PIR sensor triggers whenever a person comes in front of it within its range. (Refer datasheet for full details.) If no movement is observed in front of the sensor for a long time, the relay may get switched off. So time delay is adjusted with 1-mega-ohm variable resistor (VR1). Please read Construction and Testing section of the article for details.

MOTOR AND DRIVE SYSTEM

I read 'Selecting an Electric Motor and Drive System' buyers' guide article published in December 2016 issue with great interest. However, the speeds indicated for AC motors on page 82 may need correction. 3600rpm (2-pole), 1800rpm (4-pole) and 1200rpm (6-pole) are applicable to 60Hz line frequency.

For 50Hz line-frequency supply available in India, these should be 3000rpm (2-pole), 1500rpm (4-pole) and 1000rpm (6-pole). This is not a big error, if line frequency is mentioned.

V. Ramprakash
Madurai

EFY. Yes, line frequency (60Hz) was missing in the article. Thanks for pointing out the mistake!